

Homework 5

Ans 1 (a) $\sin \alpha = \frac{b}{R} \Rightarrow b = R \sin \alpha$

$$\theta + 2\alpha = \pi \Rightarrow \alpha = \frac{\pi - \theta}{2}$$

$$\begin{aligned} \Rightarrow b &= R \sin \{(\pi - \theta)/2\} \\ &= R \cos(\theta/2) \end{aligned}$$

(b) $D(\theta) \equiv d\sigma/d\Omega$

$$d\sigma = b db d\phi$$

$$d\Omega = \sin \theta d\theta d\phi$$

$$D(\theta) = \frac{b}{\sin \theta} \left| \frac{db}{d\theta} \right|$$

now $\frac{db}{d\theta} = -\frac{R}{2} \sin(\theta/2)$

$$\Rightarrow D(\theta) = \cancel{R} \frac{R \cos(\theta/2)}{\sin \theta} \frac{R}{2} \sin(\theta/2)$$

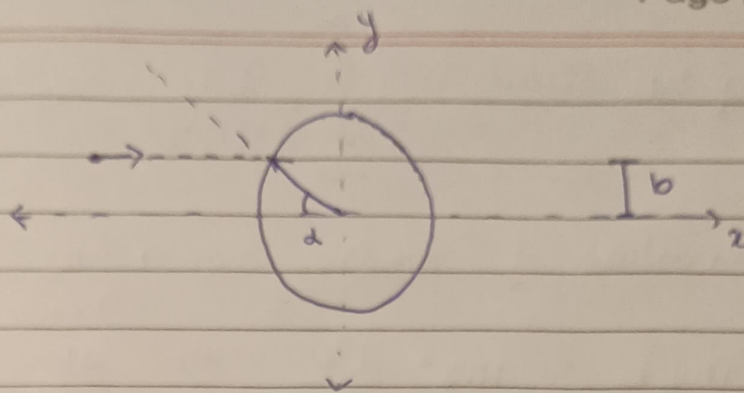
$$\Rightarrow \boxed{D(\theta) = R^2/4}$$

(c) $\sigma_T = \int D(\theta) d\Omega = \int_0^{2\pi} \int_0^\pi D(\theta) \sin \theta d\theta d\phi$

$$D(\theta) = R^2/4 : \quad \sigma_T = \frac{R^2}{4} \int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta$$

$$\Rightarrow \sigma_T = \frac{R^2 \pi}{2} \int_0^\pi \sin \theta d\theta = \pi R^2$$

Ans 2 - (a)



$$b = R \sin \alpha$$

$$\phi + 2\alpha = \pi \Rightarrow \alpha = \frac{\pi - \phi}{2}$$

$$\Rightarrow b = R \cos(\phi/2)$$

$$(b) \quad D(\phi) \equiv |dp| d\phi = |-(R/2) \sin(\phi/2)| \\ = \frac{R}{2} \sin\left(\frac{\phi}{2}\right)$$

$$(c) \quad \sigma_T = \int_0^{2\pi} D(\phi) d\phi = \frac{R}{2} \int_0^{2\pi} \sin(\phi/2) d\phi \\ = 2R$$

$$\text{Ans 3 - (a)} \quad |p_i| = |p_f| = |p| = mv_0$$

$$|\Delta p| = |p_f - p_i| = 2p \sin(\theta/2)$$

According to Newton's second law, change in momentum equals to the impulse delivered by Coulomb force

$$F = (4\pi\epsilon_0)^{-1} (q_1 q_2 / r^2)$$

$$|\Delta p| = \int F_{\text{sym}} dt$$

where F_{sym} is the component of the Coulomb force along the axis of symmetry. $F_{\text{sym}} = F \cos \psi$

$$\Rightarrow 2p \sin(\theta/2) = \int_{-\infty}^{\infty} \frac{q_1 q_2}{4\pi\epsilon_0 r^2} \cos\psi \, dt$$

by conservation of angular momentum L :

$$L = mv_\infty b = m r^2 \frac{d\psi}{dt} \Rightarrow \frac{dt}{r^2} = \frac{m}{L} d\psi$$

now, by impulse equation:

$$2p \sin(\theta/2) = \int_{-\psi_{\max}}^{\psi_{\max}} k \cos\psi \left(\frac{m}{L}\right) d\psi = \frac{km}{L} \int_{-\psi_{\max}}^{\psi_{\max}} \cos\psi \, d\psi$$

where $k = q_1 q_2 / 4\pi\epsilon_0$.

and $\psi_{\min}, \psi_{\max} = \pm \{(\pi - \theta)/2\}$

$$\Rightarrow 2mv_\infty \sin(\theta/2) = \frac{2km}{mv_\infty b} \cos\left(\frac{\theta}{2}\right)$$

$$\Rightarrow v_\infty^2 \sin(\theta/2) = \frac{k}{mb} \cos\left(\frac{\theta}{2}\right)$$

$$E = \frac{1}{2} m v_\infty^2 \Rightarrow m v_\infty^2 = 2E$$

$$\Rightarrow \boxed{b = \frac{q_1 q_2}{8\pi\epsilon_0 E} \cot\left(\frac{\theta}{2}\right)}$$

$$(b) \quad D(\theta) = \frac{b}{\sin \theta} \left| \frac{db}{d\theta} \right|$$

$$\frac{db}{d\theta} = \frac{-q_1 q_2}{16 \pi \epsilon_0 E \sin^2(\theta/2)}$$

$$\Rightarrow D(\theta) = \frac{q_1 q_2^2 \cos(\theta/2) \sin^{-2}(\theta/2)}{2 \sin(\theta/2) \cos(\theta/2) \sin(\theta/2) 8(\pi \epsilon_0 E)^2 16}$$

$$\Rightarrow D(\theta) = \left[\frac{q_1 q_2}{16 \pi \epsilon_0 E \sin^2(\theta/2)} \right]^2$$

$$(c) \quad \sigma_T = \int D(\theta) d\Omega = \int_0^{2\pi} d\phi \int_0^\pi D(\theta) \sin \theta d\theta$$

$$\Rightarrow \sigma_T = 2\pi \left(\frac{q_1 q_2}{16 \pi \epsilon_0 E} \right)^2 \int_0^\pi \frac{\sin \theta}{\sin^2(\theta/2)} d\theta$$

$$\sin \theta = 2 \sin(\theta/2) \cos(\theta/2)$$

$$\Rightarrow \sigma_T = 4\pi \left(\frac{q_1 q_2}{16 \pi \epsilon_0 E} \right)^2 \int_0^\pi \frac{\cos(\theta/2)}{\sin^3(\theta/2)} d\theta$$

σ_T diverges at $\theta \rightarrow 0$. This happens because Coulomb potential $V(r) \propto 1/r$ has infinite range.

Ans 4- (a) $\psi(r) = \sum_{l=0}^{\infty} i^l (2l+1) P_l(\cos \theta) [a_l j_l(kr) + b_l n_l(kr)]$

$$j_l(kr) \sim \frac{\sin(kr - l\pi/2)}{kr} = \frac{e^{i(kr - l\pi/2)} - e^{-i(kr - l\pi/2)}}{2ikr}$$

$$n_l(kr) \sim \frac{-\cos(kr - l\pi/2)}{kr} = -\frac{e^{i(kr - l\pi/2)} + e^{-i(kr - l\pi/2)}}{2kr}$$

$$\Rightarrow j_l \sim \frac{e^{ikr} i^{-l} - e^{-ikr} i^l}{2ikr}$$

$$n_l \sim \frac{e^{ikr} i^{-l} + e^{-ikr} i^l}{2kr} = \frac{e^{ikr} i^{-l} + e^{-ikr} i^l}{2ikr \cdot i}$$

substituting into $\psi(\vec{r})$:

$$\psi(\vec{r}) \sim \sum_{l=0}^{\infty} \frac{(2l+1)}{2i} P_l(\cos \theta) \left[(a_l - ib_l) \frac{e^{ikr}}{kr} + (-1)^{l+1} (a_l + ib_l) \frac{e^{-ikr}}{kr} \right]$$

(b) $\psi(R) = 0 \Rightarrow$ that phase shift of δ_l will occur.

$$a_l - ib_l = e^{2i\delta_l} \quad a_l + ib_l = 1$$

$$\Rightarrow 2a_l = e^{2i\delta_l} + 1 = 2e^{i\delta_l} \cos \delta_l$$

$$\Rightarrow a_l = e^{i\delta_l} \cos \delta_l$$

$$\Rightarrow \psi(\vec{r}) = \sum_{l=0}^{\infty} i^l (2l+1) P_l(\cos \theta) e^{i\delta_l} [\cos \delta_l j_l(kr) - \sin \delta_l n_l(kr)]$$

(c) $\psi(R) = 0$

$$\cos \delta_l j_l(kR) - \sin \delta_l n_l(kR) = 0$$

$$\Rightarrow \sin \delta_l j_l(kR) = (\cos \delta_l) n_l(kR)$$

$$\Rightarrow \boxed{\tan \delta_l = \frac{j_l(kR)}{n_l(kR)}}$$

(d) $l=0$: $j_0(x) = \frac{\sin x}{x}$ $n_0(x) = -\frac{\cos x}{x}$

$$\Rightarrow \tan \delta_0 = -\tan(kR) \Rightarrow \delta_0 = -kR$$

$$l=1 : j_1(x) = \frac{\sin x}{x^2} - \frac{\cos x}{x}$$

$$n_1(x) = -\frac{\cos x}{x^2} - \frac{\sin x}{x}$$

$$l=2 : j_2(x) = \left(\frac{3}{x^3} - \frac{1}{x} \right) \sin x - \frac{3}{x^2} \cos x$$

$$n_2(x) = -\left(\frac{3}{x^3} - \frac{1}{x} \right) \cos x - \frac{3}{x^2} \sin x$$

l	$\tan \delta_l$ for $kR = \pi/6$	$\tan \delta_l$ for $kR = \pi/4$
0	$-\pi/6$	$-\pi/4$
1	-0.0417	-0.1202
2	-0.0068	-0.0057

(e)

$$kR \ll 1$$

$$\tan \delta_l = \frac{j_l(kR)}{n_l(kR)} \approx \left\{ \frac{(kR)^l}{(2l+1)!!} \right\} \left\{ \frac{-(2l-1)!!}{(kR)^{l+1}} \right\}$$

$$\Rightarrow \tan \delta_l \approx \frac{-(kR)^{2l+1}}{(2l+1)!! (2l-1)!!}$$

$$\text{for } kR \ll 1: \quad \delta_l \approx \frac{-(kR)^{2l+1}}{(2l+1)!! (2l-1)!!}$$

(f)

$$\sigma_T = \frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) \sin^2 \delta_l$$

(i) For $kR = \pi/6$:

$$\sum_{l=0}^2 (2l+1) \sin^2 \delta_l \approx 0.2551$$

$$\sigma_T = \frac{4\pi (0.2551)}{k^2} \approx 3.72 \pi R^2$$

(ii) $kR = \pi/4$

$$\sum_{l=0}^2 (2l+1) \sin^2 \delta_l \approx 0.5429$$

$$\sigma_T = \frac{4\pi (0.5429)}{k^2} \approx 3.52 \pi R^2$$

$$(g) \quad \sin^2 \delta_0 = \sin^2(kR) \approx \frac{(kR)^2 - (kR)^4}{3}$$

$$\delta_1 \approx \tan \delta_1 \approx -\frac{(kR)^3}{3}$$

$$\sin^2 \delta_1 \approx \left(\frac{-k^3 R^3}{3}\right)^2 = \frac{(kR)^6}{9}$$

$$\sigma_T = \frac{4\pi}{k^2} [\sin^2 \delta_0 + 3\sin^2 \delta_1 + \dots]$$

$$= \frac{4\pi}{k^2} [(kR)^2 - \frac{1}{3}(kR)^4 + O(kR)^6]$$

$$= 4\pi R^2 \left[1 - \frac{1}{3}(kR)^2 + O(kR)^4 \dots \right]$$

$$\Rightarrow \boxed{\sigma_T = 4\pi R^2 \left[1 - \frac{(kR)^2}{3} + \dots \right]}$$

Ans 5- $\sigma_T = 4\pi R^2 \left[1 - \frac{\tan(qR)}{qR} \right]^2$

$$\frac{-\hbar^2}{2m} \frac{d^2 u(r)}{dr^2} + V(r)u(r) = E u(r)$$

potential well: $-V_0, \quad r \leq R$
 $0, \quad r > R$

inside: $\frac{d^2 u_1}{dr^2} + q^2 u_1 = 0, \quad q^2 = \frac{2m(E+V_0)}{\hbar^2}$

$$u_{\text{in}}(r) = A \sin(qr) + B \cos(qr)$$

$$\psi(r) = u(r)/r \quad \text{must be finite at } r=0$$

$$\Rightarrow u_{\text{in}}(0) = 0$$

$$\Rightarrow B = 0$$

$$\Rightarrow u_{\text{in}}(r) = U \sin(qr)$$

$$\text{for } r > R: \frac{d^2 u_{\text{out}}}{dr^2} + k^2 u_{\text{out}} = 0, \quad k^2 = \frac{2mE}{\hbar^2}$$

$$\Rightarrow u_{\text{out}}(r) = C \sin(kr + \delta_0)$$

$$\text{Boundary cond: } u_{\text{out}}(R) = u_{\text{in}}(R)$$

$$u'_{\text{out}}(R) = u'_{\text{in}}(R)$$

$$\Rightarrow \frac{qA \cos(qR)}{A \sin(qR)} = \frac{kC \sin(kR + \delta_0)}{C \sin(kR + \delta_0)}$$

$$\Rightarrow q \cot(qR) = k \cot(kR + \delta_0)$$

in low energy limit, $k \rightarrow 0 \Rightarrow \delta_0 \rightarrow \text{small}$

$$k \cot(kR + \delta_0) = \frac{k \cos(kR + \delta_0)}{\sin(kR + \delta_0)}$$

$$\approx \frac{k}{kR + \delta_0}$$

$$\Rightarrow q \cot(qR) = \frac{k}{kR + \delta_0}$$

$$\Rightarrow S_0 = k \left(\frac{\tan(qR)}{q} - R \right)$$

$$\sigma_T = \frac{4\pi}{k^2} \sin^2(S_0), \quad S_0 \text{ is small}$$

$$\Rightarrow \sigma_T = 4\pi \left(R - \frac{\tan(qR)}{q} \right)^2$$

$$\Rightarrow \sigma_T = 4\pi R^2 \left[1 - \frac{\tan(qR)}{qR} \right]^2$$

Ans 6- (a) for $l=0$: $-\frac{\hbar^2}{2m} u''(r) + V_0 \delta(r-R)u(r) = Eu(r)$

$$k^2 = 2mE/\hbar^2 : u''(r) + k^2 u(r) = \frac{2mV_0}{\hbar^2} \delta(r-R)u(r)$$

$$r < R : u(r) = A \sin kr$$

$$r > R : u(r) = B \sin(kr + S_0)$$

$$u_L(R) = u_R(R) \Rightarrow A \sin(kR) = B \sin(kR + S_0)$$

discontinuity in 1st derivative:

$$u'_>(R) - u'_<(R) = \frac{2mV_0}{\hbar^2} u(R)$$

$$P = 2mV_0 R/\hbar$$

$$\Rightarrow k B \cos(kR + S_0) - k A \cos(kR) = \frac{P}{R} A \sin(kR)$$

$$k \cot(kR + \delta_0) - k \cot(kR) = \beta / R$$

$$\text{let } x = kR, \Rightarrow \cot(x + \delta_0) = \frac{\beta}{x} + \cot x$$

$$\Rightarrow \frac{-\sin \delta_0}{(\sin x) \sin(x + \delta_0)} = \frac{\beta}{x}$$

$$\Rightarrow -x \sin \delta_0 = \beta \sin x \sin(x + \delta_0)$$

$$\Rightarrow \cancel{\cot} -x = \beta \sin x [\cos x + \cot \delta_0 \sin x]$$

$$\Rightarrow \cot \delta_0 = - \frac{(x + \frac{\beta}{2} \sin(2x))}{\beta \sin^2 x}$$

$$\text{now } \sin^2(\delta_0) = \frac{1}{1 + \cot^2(\delta_0)}$$

$$\Rightarrow \cot^2(\delta_0) = \frac{\left\{ x + \frac{\beta}{2} \sin(2x) \right\}^2}{\beta^2 \sin^4 x}$$

$$\Rightarrow 1 + \cot^2 \delta_0 = \frac{\beta^2 \sin^4 x + x^2 + x\beta \sin 2x + \frac{\beta^2}{4} \sin^2 2x}{\beta^2 \sin^4 x}$$

$$\begin{aligned} \beta^2 \sin^4 x + \beta^2 \sin^2 x \cos^2 x &= \beta^2 \sin^2 x \{ \sin^2 x + \cos^2 x \} \\ &= \beta^2 \sin^2 x \end{aligned}$$

$$\Rightarrow \text{Num} = x^2 + x\beta \sin(2x) + \beta^2 \sin^2 x$$

$$\Rightarrow \sin^2 \delta_0 = \frac{\beta^2 \sin^4 x}{x^2 + x\beta \sin 2x + \beta^2 \sin^2 x}$$

(b) let $x = kR \ll 1 \Rightarrow \sin x \approx x$

$$\Rightarrow \sin^2 \delta_0 = \frac{\beta^2 x^4}{x^2 + x^2(2\beta) + \beta^2 x^2}$$

$$= \frac{\beta^2 x^2}{1 + \beta^2 + 2\beta} = \frac{\beta^2 x^2}{(1+\beta)^2}$$

$$\sigma_T = \frac{4\pi}{k^2} \sin^2 \delta_0 = \frac{4\pi}{k^2} \frac{\beta^2 k^2 R^2}{k^2 (1+\beta)^2}$$

$$\Rightarrow \boxed{\sigma_T = 4\pi R^2 \left[\frac{\beta}{1+\beta} \right]^2}$$

Ans 7) $q(\vec{r}, \vec{r}') = \int \frac{d^3 k'}{(2\pi)^3} \frac{e^{i\vec{k}' \cdot (\vec{r} - \vec{r}')}}{k^2 - (k')^2 \pm i\epsilon}$

$\vec{R} = \vec{r} - \vec{r}'$ and $R = |\vec{R}|$

switch to spherical coordinates for k' , where polar angle θ is measured relative to $\vec{R}$

$$q(R) = \frac{1}{(2\pi)^3} \int_0^\infty (k')^2 dk' \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\phi \frac{e^{i\vec{k}' \cdot \vec{R} \cos\theta}}{k^2 - (k')^2 \pm i\epsilon}$$

$$= \frac{1}{(2\pi)^3} (2\pi) \int_0^\infty (k')^2 dk' \frac{2\sin(k'R)}{k'R} \frac{1}{k^2 - (k')^2 \pm i\epsilon}$$

$$= \frac{1}{2\pi^2 i R} \int_0^\infty \frac{k' \sin(k'R) dk'}{k^2 - (k')^2 \pm i\epsilon}$$

$$= \frac{1}{2\pi^2 i R} \int_{-\infty}^{\infty} \frac{k' e^{i k' R} dk'}{2i(k^2 - (k')^2 \pm i\epsilon)}$$

$$= \frac{-1}{4\pi^2 i} \int_{-\infty}^{\infty} \frac{k' e^{i k' R} dk'}{\{k' - (k \pm i\epsilon)\} \{k' + (k \pm i\epsilon)\}}$$

By Cauchy's residue theorem:

$$Q(\vec{r}, \vec{r}') = \frac{-1}{4\pi} \frac{e^{\pm i k |\vec{r} - \vec{r}'|}}{|\vec{r} - \vec{r}'|}$$

Ans 8) $\left[\frac{-\hbar^2}{2m} \nabla^2 + V(\vec{r}) \right] \psi(\vec{r}) = E \psi(\vec{r})$

$$E = \hbar^2 k^2 / 2m \Rightarrow (\nabla^2 + k^2) \psi(\vec{r}) = \frac{2m}{\hbar^2} V(\vec{r}) \psi(\vec{r})$$

general solⁿ: $\psi(\vec{r}) = \psi_0(\vec{r}) + \int Q(\vec{r}, \vec{r}') f(\vec{r}') d^3 r'$

the solⁿ to homogeneous eqⁿ is $\psi_0(\vec{r}) = e^{i \vec{k} \cdot \vec{r}}$

now substitute $Q(\vec{r}, \vec{r}')$ and $f(\vec{r}')$:

$$\psi(\vec{r}) = e^{i \vec{k} \cdot \vec{r}} + \int \left(\frac{-1}{4\pi} \frac{e^{i k |\vec{r} - \vec{r}'|}}{|\vec{r} - \vec{r}'|} \right) \left(\frac{2m}{\hbar^2} V(\vec{r}') \psi(\vec{r}') \right) d^3 r'$$

$$\Rightarrow \psi(\vec{r}) = e^{i \vec{k} \cdot \vec{r}} + \frac{2m}{\hbar^2} \int d^3 r' \left[\frac{-e^{i k |\vec{r} - \vec{r}'|}}{4\pi |\vec{r} - \vec{r}'|} \right] V(\vec{r}') \psi(\vec{r}')$$

Ans 9)
$$f_B(\theta) = \frac{-m}{2\pi\hbar^2} \int d^3r' e^{i\vec{q}\cdot\vec{r}'} V(\vec{r}')$$

for spherically symmetric potentials :

$$f_B(\theta) = \frac{-2m}{\hbar^2} \int_0^\infty r V(r) \sin(qr) dr$$

$$q = |\vec{k}_s - \vec{k}| = 2k \sin(\theta/2)$$

(a) (1) $V(\vec{r}) = V_0 \lambda^3 \delta^3(\vec{r})$

$$\begin{aligned} f_B(\theta) &= \frac{-m}{2\pi\hbar^2} \int d^3r' e^{i\vec{q}\cdot\vec{r}'} [V_0 \lambda^3 \delta^3(\vec{r}')] \\ &= \frac{-m V_0 \lambda^3}{2\pi\hbar^2} \end{aligned}$$

(2) $V(\vec{r}) = V_0 \lambda^5 \nabla^2 \delta^3(\vec{r})$

$$\int d^3r' e^{i\vec{q}\cdot\vec{r}'} \nabla^2 \delta^3(\vec{r}') = \int d^3r' \delta^3(\vec{r}') \nabla^2 (e^{i\vec{q}\cdot\vec{r}'})$$

$$\nabla^2 e^{i\vec{q}\cdot\vec{r}'} = (i\vec{q}) \cdot (i\vec{q}) e^{i\vec{q}\cdot\vec{r}'} = -q^2 e^{i\vec{q}\cdot\vec{r}'}$$

$$\int d^3r' \delta^3(\vec{r}') (-q^2 e^{i\vec{q}\cdot\vec{r}'}) = -q^2$$

$$f_B(\theta) = \frac{-m}{2\pi\hbar^2} (V_0 \lambda^5) (-q^2) = \frac{m V_0 \lambda^5 q^2}{2\pi\hbar^2}$$

$$(3) \int d^3r e^{i\vec{q}\cdot\vec{r}} e^{-\alpha r^2} = \left(\frac{\pi}{\alpha}\right)^{3/2} e^{-q^2/4\alpha}$$

where $\alpha = 1/4\lambda^2$

$$\int d^3r e^{i\vec{q}\cdot\vec{r}} e^{-r^2/(4\lambda^2)} = (4\pi\lambda^2)^{3/2} e^{-q^2\lambda^2} = 8\pi \lambda^3 e^{-q^2\lambda^2}$$

$$f_B(\theta) = \frac{-m}{2\pi\hbar^2} V_0 (8\pi \lambda^3 e^{-q^2\lambda^2}) = \frac{-4mV_0 \lambda^3 \sqrt{\pi}}{\hbar^2} e^{-q^2\lambda^2}$$

$$(b) (i) f_B(\theta) = \frac{-2mV_0}{\hbar^2 q} \int_0^R r \sin(qr) dr$$

integrate by parts

$$\int_0^R r \sin(qr) dr = \frac{\sin(qR) - qR \cos(qR)}{q^2}$$

$$\Rightarrow f_B(\theta) = \frac{-2mV_0}{\hbar^2 q} \left[\frac{\sin qR - qR \cos qR}{q^2} \right]$$

for $kR = 1/2$, $q = 2k \sin(\theta/2)$
 $= \sin(\theta/2)$

zeros of $f_B(\theta)$: $\tan qR = qR$

happens when $qR = 0, 4.49, \dots$
 but $1 < 4.49$,
 no valid scattering θ is possible

$$(ii) \sigma_T = \int |f_B(\theta)|^2 d\Omega = \left(\frac{2mV_0 R^3}{\hbar^2} \right)^2 \frac{2\pi}{k^2 R^2} I$$

$$I = \int_0^d \frac{(\sin x - x \cos x)^2}{x^5} x dx$$

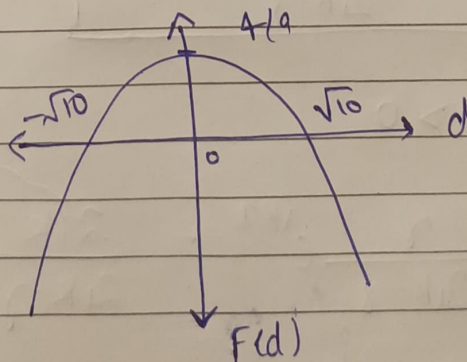
$$\Rightarrow \sigma_T = \pi R^2 \left[\frac{2mV_0 R^2}{\hbar^2} \right]^2 \frac{8}{d^2} \int_0^d \frac{(\sin x - x \cos x)^2}{x^5} dx$$

$$\Rightarrow \sigma_T = \pi R^2 \left[\frac{2mV_0 R^2}{\hbar^2} \right]^2 \frac{(2d^4 - 2d^2 + 2d \sin 2d - 1 + \cos 2d)}{d^6}$$

(iii) ~~as $d \rightarrow 0$: F~~

$$\Rightarrow \sigma_T = \pi R^2 \left[\frac{2mV_0 R^2}{\hbar^2} \right]^2 F(d)$$

$$(iii) \text{ as } d \rightarrow 0, F(d) \rightarrow \frac{4}{9} - \frac{2d^2}{45} + \dots$$



$$(iv) F(d)_{\max} = 4/9$$

$$\sigma_T^{\max} = \pi R^2 \left(\frac{2mV_0 R^2}{\hbar^2} \right)^2 \frac{4}{9} = \boxed{\pi R^2 \left(\frac{4mV_0 R^2}{3\hbar^2} \right)^2}$$

Ans 10) $\psi(x) = \phi_0(x) + \int_{-\infty}^{\infty} G(x, x') U(x') \psi(x') dx'$

$$V(x) = \frac{\hbar^2}{2m} \gamma \delta(x) \quad , \quad U(x) = \gamma \delta(x)$$

$$\Rightarrow \psi(x) = e^{ikx} + \frac{\gamma}{2ik} e^{i|k|x} \psi(0)$$

$$\psi(0) = e^0 + \frac{\gamma}{2ik} \psi(0) \quad \Rightarrow \quad \cancel{\psi(0)} = 1 + \frac{\gamma}{2ik} \cancel{\psi(0)}$$

$$\Rightarrow \boxed{\psi(0) = \frac{2ik}{2ik - \gamma}}$$

$$\psi(x) = e^{ikx} + \left(\frac{\gamma}{2ik} \right) \left(\frac{2ik}{2ik - \gamma} \right) e^{i|k|x}$$

$$= \boxed{e^{ikx} + \frac{\gamma}{2ik - \gamma} e^{i|k|x}}$$

Now for $x > 0$

$$\psi(x) = e^{ikx} + \frac{\gamma}{2ik - \gamma} e^{ikx} = \frac{2ik}{2ik - \gamma} e^{ikx}$$

$$\Rightarrow \boxed{t = \frac{2ik}{2ik - \gamma}}$$

For $x < 0$, $|x| = -x$

$$\Rightarrow \psi(x) = e^{ikx} + \frac{\gamma}{2ik - \gamma} e^{-ikx} \Rightarrow \boxed{r = \frac{\gamma}{2ik - \gamma}}$$

$$T = \frac{4k^2}{4k^2 + \gamma^2}$$

$$R = \frac{\gamma^2}{4k^2 + \gamma^2}$$

$$\text{Ans 11)} \quad T_{fi} = \frac{2\pi}{\hbar} \int_{p_f=0}^{\infty} |V_{fi}|^2 \delta\left(\frac{p_f^2}{2m} - \frac{p_i^2}{2m}\right) \frac{L^3 p_f^2 dp_f d\Omega}{(2\pi\hbar)^3}$$

$$\text{let } E_f = \frac{p_f^2}{2m} \Rightarrow dE_f = \frac{p_f}{m} dp_f$$

$$\Rightarrow T_{fi} = \frac{2\pi}{\hbar} \int |V_{fi}|^2 \delta(E_f - E_i) \frac{L^3 p_f^2 d\Omega}{(2\pi\hbar)^3 p_f} dE_f$$

$$\Rightarrow \frac{dT_{fi}}{d\Omega} = \frac{2\pi}{\hbar} |V_{fi}|^2 \frac{L^3 m p_i}{(2\pi\hbar)^3}$$

$$V_{fi} = \langle f | V(r) | i \rangle = \frac{1}{L^3} \int d^3r e^{i(k_i - k_f) \cdot r} V(r)$$

$$f_B(0) = \frac{-m}{2\pi\hbar^2} \int d^3r e^{i(k_i - k_f) \cdot r} V(r)$$

$$\Rightarrow V_{fi} = \frac{1}{L^3} \left(\frac{-2\pi\hbar^2}{m} f_B(0) \right)$$

$$\Rightarrow |V_{fi}|^2 = \frac{1}{L^6} \frac{4\pi^2 \hbar^4}{m^2} |f_B(0)|^2$$

substituting into integral :

$$\frac{dT_f}{d\Omega} = \frac{2\pi}{\hbar} \left[\frac{1}{L^3} \frac{4\pi^2 \hbar^4}{m^2} |f_B(\theta)|^2 \right] \frac{L^3 m P_i}{(2\pi\hbar)^3}$$

$$\Rightarrow \frac{dT_f}{d\Omega} = \frac{P_i}{mL^3} |f_B(\theta)|^2$$

$$j_{in} = P_i / mL^3, \quad P_i = P_f \text{ for elastic scattering}$$

$$\Rightarrow \boxed{\frac{dT_f}{d\Omega} = j_{in} |f_B(\theta)|^2}$$

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{1}{j_{in}} \frac{dT_f}{d\Omega} = |f_B(\theta)|^2}$$

b